

Summer Internship – VISA Inc. '23

About VISA :

VISA is an American Multinational Company and a leading digital payment technology firm. It's headquartered in California, USA. VISA's often miscomprehended as a debit/credit card issuer, but its not what it does and the way it actually makes money. It facilitates money transfers between bank accounts through VISA branded cards.

The hiring process of VISA included 2 rounds :

- i. Online Test
- ii. Interview (Technical cum Managerial)

Online test: (Platform – HackerRank)

We had two questions which were supposed to be completed within an hour. The difficulty of the questions was moderate. Each one got different set of questions.

The questions I got :

1. *Odd One Out*

Given an array of strings, find the odd one out by analyzing the difference in adjacent letters of the string.

Eg. ["abcd", "wxyz", "pqrs", "acdf"]

The odd one out here is "acdf".

As the difference between two consecutive letters of "abcd", "wxyz" and "pqrs" is same i.e, 1. But considering "acdf", 'c'-'a'=2, 'd'-'c'=1, 'f'-'d'=2.

2. *Minimum Cost – Jump Game*

Given an 1-indexed array of nums and a value 'k'. You start from index-0. In one move, you can jump at most 'k' steps forward without going outside the boundaries of the array. You want to reach the last index of the array (index n). Your **score** is the **sum** of all nums[i] for each index 'i' you visited in the array.

Return the **minimum score** you can get.

Similar to [Jump Game VI - LeetCode](#).

We were asked just to fill the given function.

I could solve both the questions completely. 14 students were shortlisted for the next round.

Tips :

- Be consistent in solving coding questions in platforms like LeetCode, GeeksForGeeks, HackerRank etc.,
- Try understanding the concepts and underlying patterns. Don't memorize soln.
- Follow ACM's C.O.D.E program, if you've no idea on how to plan or go on with. The curated lists of problems covering almost all possible patterns will really be helpful.

- Also try to practice coding the given problem from scratch including the driver code (rather just filling the given function as in LeetCode) as several companies would expect you to do so.
- Time matters the most. Try completing the test as early as possible. Sometimes even brute force approach would suffice to pass all the test cases. (NOTE: Try Brute Force Approach only if you can't come up with an optimised solution)

Interview: (Platform – Microsoft Teams)

My interview was scheduled at 10:00 a.m. But the interviewer who was actually assigned couldn't make it on time. I was waiting for more than an hour. Finally, my interview got rescheduled to 11:45 a.m, with another manager.

He first asked me to introduce myself, which's a typical icebreaker in every 1-1 interview. *(Be cautious about what you say and also try to cover each and every side of you.)*

He then moved to the code-pair. Usually you'll be given link for code pair, where you'd be asked to solve a question along with interviewer and this would mostly be in HackerRank. I was also given one. But, as the interviewer came unprepared (he told that he had got finalised just then), he asked me that did you remember the problem you had got in your online test, which I answered yes. He asked me to explain the problem statement and the approach I had used to solve. So, I explained my approach for both the questions. He then asked for an optimised/alternate solutions.

For the first question, I solved it with $O(n*s)$ time (n-size of the array and s-length of the string) and $O(n)$ space. After explaining this approach, I gave a space optimised approach.

For the second question, I solved using dynamic programming. I then explained memoised and tabulated approach and a greedy solution as an alternative.

He was satisfied with the given approaches. He gave me 2-3 sample test cases to check my solution and my solution could pass the given cases. I wasn't asked to code.

Questions followed:

- i. Do you know Java?
- ii. Other subjects offered apart from Java? (As I answered no, we have it as an elective in the upcoming semester for the prev. qn.) And why would you choose Java?
- iii. Why intern at VISA?
- iv. Do you know what VISA does?
- v. How VISA's getting revenue?

The last two questions lead us to a discussion for more than 15-20 minutes, where I explained in detail the business model, what company does, how the company actually makes money etc., where he added on some points now and then. I was able to answer these questions as I intensively researched about the company and also I got a few points from my seniors.

He finally asked whether I had any questions for him and I asked 3-4 questions. And it was more kinda discussion than a Q&A. That's the end. The interviewer was very friendly to talk to and he made me so comfortable. The interview lasted for an hour.

Preparation: (Based on how I prepared)

- For code-pair, the preparation you've done for first round is enough.
- DSA is super-important. Have a good understanding of every data structure and algorithm.
- Refer GeeksForGeeks for DSA theory and concepts, for extra conceptual clarity – YouTube channels like Abdul Bhari, Tech Dose, NeetCode etc., and explore the Discuss section in LeetCode for excellent articles on algorithmic topics like DP, Backtracking, Binary Search etc.,
- For OS, GFG's last minute notes will be sufficient and for DBMS, class notes will do and for additional info refer GFG articles and Javatpoint.
- For OOPs, cover the topics in Fraz's OOP sheet and for concepts refer GFG and Codaming - VeDinesh Academy in YouTube (Complete OOP in C++).
- Have 2-3 solid projects and be ready to answer any question that could possibly be posted.
- Although VISA don't ask aptitude questions in the online test, it is equally important as the other subjects, as several other companies ask for. Refer IndiaBix for the same.

Note: My interview experience might be quite misleading, as I wasn't posted any technical questions from OS, DBMS, OOPs and my projects. Though I was very well prepared on these topics expecting for the worst case scenario, I didn't fortunately get such questions. But it might not be the same way as I had. ***So, do prepare these topics well.***

General Take Aways:

- Know your resume well. Add only what you know. The interviewer will only check for how good your understanding level is, in the things you've mentioned. Don't lie in your resume.
- Research about the company the day before your interview very well, which helped me a lot in my case, as I was posted questions based only on the company.
- Be confident while answering, and cautious too because you're the one who's gonna drive the interview. Don't keep mum while solving question during code-pair.
- Refer the interview archives of the company, on the previous day of the interview and talk to your seniors to get some ideas.
- Consistency is the key. And start your preparation atleast 2 months prior to the intern drive (But starting earlier will make your preparation journey a bit stress-free). Also, it's never too late to start your preparation, if you're short of time. You can still do well with a lil' extra effort. Have a strong mindset! Have ***small*** breaks and relaxations, every now and then.

Wishing you all the very best for your internship drive. ✨

Feel free to contact 📞

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